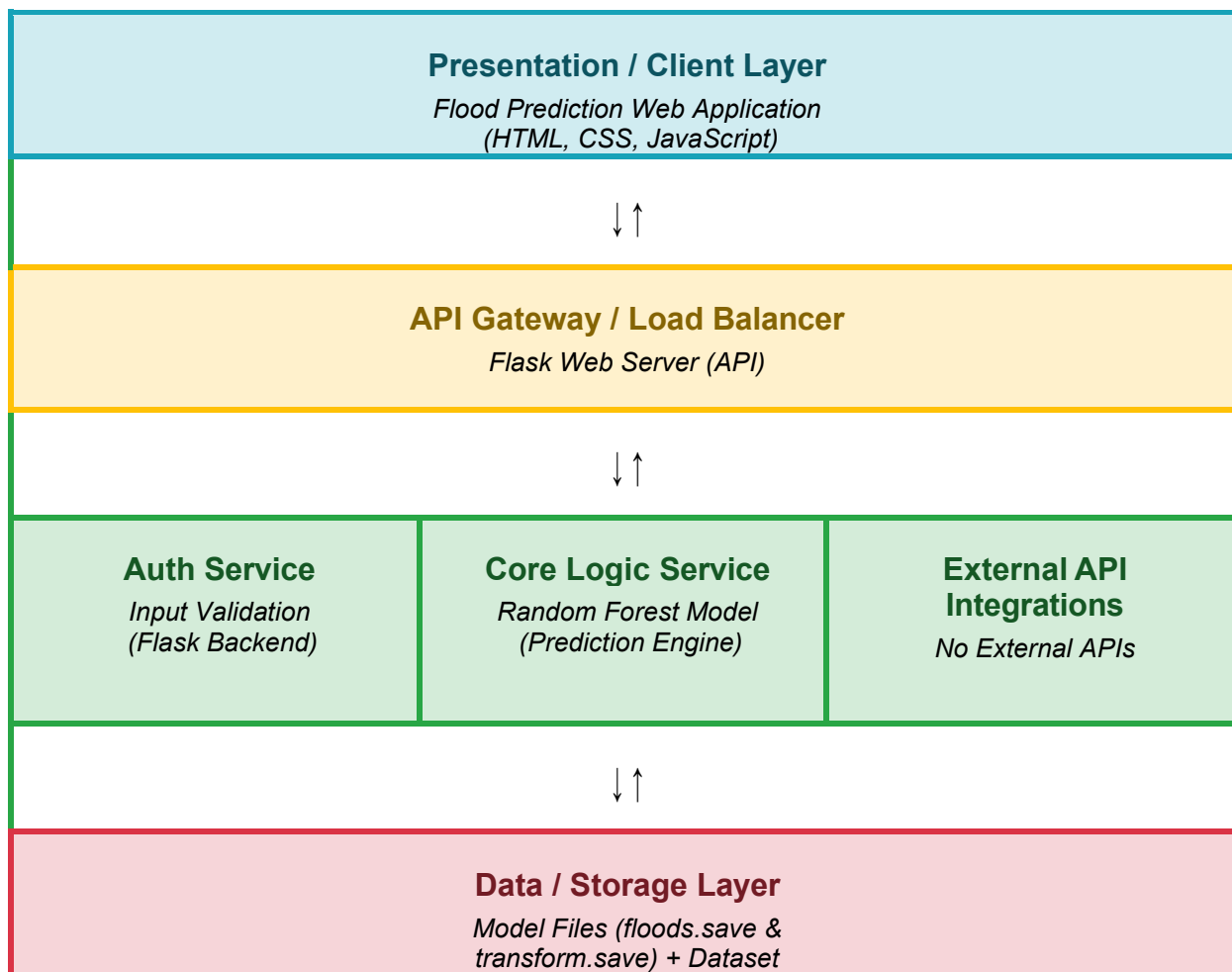


Solution Architecture

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where users enter weather and rainfall parameters and view prediction results.	HTML5, CSS3, JavaScript
API Gateway	Flask receives HTTP requests from the web interface, validates input, and forwards the data to the machine learning model.	Python, Flask
Core Logic Service	Processes the user input, performs data transformation using the saved scaler, loads the trained Random Forest model, and predicts whether there is a flood chance.	Scikit-learn, Random Forest, Pickle
Data Layer	Stores the trained machine learning model (floods.save), preprocessing transformer (transform.save), and historical flood dataset used for training.	Pickle (.save), Excel Dataset (.xlsx)